(c) $$\frac{\partial}{\partial t}\int_{x-t}^{x+t} t(x+y)\,dy$$

- With Leibniz rule:

$$\frac{\partial}{\partial t}\int_{x-t}^{x+t} t(x+y)\,dy = \int_{x-t}^{x+t}\frac{\partial}{\partial t}\left[t(x+y)\right]dy +$$

$$+\,t\left[x+(x+t)\right]\frac{\partial}{\partial t}(x+t) - t\cdot\left[x+(x-t)\right]\cdot\frac{\partial}{\partial t}(x-t) =$$

$$=\int_{x-t}^{x+t}(x+y)\,dy + t\left[2x+t\right]\cdot 1 - t\left[2x-t\right]\cdot(-1) =$$

$$=\left[xy+\frac{y^2}{2}\right]_{y=x-t}^{y=x+t} + t(2x+t) + t(2x-t) =$$

$$=\left[x(x+t)+\frac{(x+t)^2}{2} - x(x-t) - \frac{(x-t)^2}{2}\right] + t\left[2x+\cancel{t}+2x-\cancel{t}\right] =$$

$$=\left[x(\cancel{x}+t-\cancel{x}+t)+\frac{1}{2}\left(\cancel{x^2}+2xt+\cancel{t^2}-\cancel{x^2}+2xt-\cancel{t^2}\right)\right]+t\cdot(4x) =$$

$$=\left[2tx+\frac{1}{\cancel{2}}\left(\overset{2}{\cancel{4}}xt\right)\right]+4xt = \left[4xt\right]+4xt = 8xt.$$

So,

(6) $$\frac{\partial}{\partial t}\int_{x-t}^{x+t} t(x+y)\,dy = 8xt.$$

Without Leibniz rule:

(7) $$\frac{\partial}{\partial t}\int_{x-t}^{x+t}(tx+ty)\,dy = \frac{\partial}{\partial t}\left[txy+\frac{ty^2}{2}\right]_{y=x-t}^{y=x+t} = \frac{\partial}{\partial t}\left[2xt^2+\frac{t}{2}(4xt)\right] = 8xt.$$

The results in (6) & (7) match each other, as desired.

(d) Find $\frac{dy}{dx}$ if

(8) $$\int_x^{2y} e^{-xt}\,dt = -2$$

Sol'n.

Consider $y = y(x)$ & using Leibniz rule, take $\frac{d}{dx}(\cdot)$ of BS of eq'n (8):

(9) $$\frac{d}{dx}\left[\int_x^{2y} e^{-xt}\,dt\right] = \underbrace{\frac{d}{dx}[-2]}_{=0}.$$ Thus

$$0 = \frac{d}{dx}\int_x^{2y} e^{-xt}\,dt = \int_x^{2y} \frac{\partial}{\partial x}\left[e^{-xt}\right]dt + e^{-x\cdot(2y)}\cdot\frac{d}{dx}(2y)$$

$$- e^{-x\cdot(x)}\cdot\frac{d}{dx}[x] = \int_x^{2y} -t\cdot e^{-xt}\,dt + e^{-2xy}\cdot 2\cdot\frac{dy}{dx} - e^{-x^2}\cdot 1 =$$

$\int\cdot dt$

Integ. by parts

$$\int_x^{2y} \underbrace{-t}_{u}\cdot \underbrace{d\left[\frac{e^{-xt}}{-x}\right]}_{dv} + 2e^{-2xy}\cdot\left(\frac{dy}{dx}\right) - e^{-x^2} =$$

$$= (-t)\cdot\left[\frac{e^{-xt}}{-x}\right]_{t=x}^{t=2y} - \int_x^{2y} \frac{e^{-xt}}{-x}\,d(-t) + 2e^{-2xy}\cdot\left(\frac{dy}{dx}\right) - e^{-x^2} =$$

$$= \frac{1}{x}\left[(2y)\cdot e^{-x\cdot(2y)} - (x)\cdot e^{-x\cdot(x)}\right] - \left[\frac{1}{x}\cdot\frac{e^{-xt}}{-x}\right]_{t=x}^{t=2y} +$$

$$+ 2e^{-2xy}\left(\frac{dy}{dx}\right) - e^{-x^2} =$$

$$= \frac{2y}{x} \cdot e^{-2xy} - e^{-x^2} + \frac{1}{x^2} \cdot \left[e^{-x \cdot (2y)} - e^{-x \cdot (x)} \right] + 2e^{-2xy} \left[\frac{dy}{dx} \right] - e^{-x^2} =$$

$$= \frac{2y}{x} \cdot e^{-2xy} - e^{-x^2} + \frac{e^{-2xy}}{x^2} - \frac{e^{-x^2}}{x^2} + 2e^{-2xy} \left[\frac{dy}{dx} \right] - e^{-x^2} \Rightarrow$$

Recall, we started the calculin by flipping the 0, zero, on the RHS to LHS, so we have:

$$0 = \left(\frac{2y}{x} + \frac{1}{x^2} \right) e^{-2xy} - \left(\frac{1}{x^2} + 2 \right) e^{-x^2} + 2e^{-2xy} \cdot \left[\frac{dy}{dx} \right] \Rightarrow$$

$$2 \cdot e^{-2xy} \left[\frac{dy}{dx} \right] = \left[\frac{1}{x^2} + 2 \right] \cdot e^{-x^2} - \left[\frac{2y}{x} + \frac{1}{x^2} \right] e^{-2xy}$$

$$\frac{dy}{dx} = \left[\frac{1}{x^2} + 2 \right] \frac{e^{-x^2}}{2e^{-2xy}} - \left[\frac{2y}{x} + \frac{1}{x^2} \right] \frac{e^{-2xy}}{2e^{-2xy}} \quad \text{or}$$

(10) $$\frac{dy}{dx} = \left[\frac{1}{2x^2} + 1 \right] e^{2xy - x^2} - \left[\frac{y}{x} + \frac{1}{2x^2} \right]$$

However, when I tried without Leibniz rule I got the following, in (11), using implicit differentian formula applied to BS of (8), I got different answer. I could not find the part which I made a mistake, but I really hope one of them is true!

$$\left(e^{-xt}\right)'_t = (-x)e^{-xt}$$

(8) $$\int_x^{2y} e^{-xt}\,dt = -2. \quad \Rightarrow$$

$$\left[\frac{e^{-xt}}{-x}\right]_{t=x}^{t=2y} = -2.$$

$$-\frac{1}{x}\left(e^{-x\cdot(2y)} - e^{-x\cdot x}\right) = -2. \quad \Big\{ \cdot \frac{d}{dx}(\cdot) \text{ of BS:}$$

$$\frac{d}{dx}\left[e^{-2xy} - e^{-x^2}\right] = \frac{d}{dx}(2x)$$

$$\frac{d}{dx}\left(e^{-2xy}\right) - \frac{d}{dx}\left(e^{-x^2}\right) = 2.$$

$$e^{-2xy}\cdot\frac{d}{dx}(-2xy) - \left[e^{-x^2}\cdot\frac{d}{dx}(-x^2)\right] = 2$$

$$e^{-2xy}\cdot(-2)\cdot\left(y\cdot\frac{dx}{dx} + x\cdot\frac{dy}{dx}\right) - \left[e^{-x^2}(-2x)\right] = 2$$

$$\frac{-2e^{-2xy}}{-2e^{-2xy}}\left(y + x\cdot\frac{dy}{dx}\right) + \frac{2\cdot x\cdot e^{-x^2}}{-2e^{-2xy}} = \frac{2}{-2e^{-2xy}}$$

$$y + x\cdot\frac{dy}{dx} - x\cdot e^{2xy-x^2} = -1\cdot e^{2xy}$$

$$\cancel{y} + x\cdot\frac{dy}{dx} = x e^{2xy-x^2} - e^{2xy} - y$$

$$\frac{1}{x}\cdot x\frac{dy}{dx} = \left(x e^{2xy-x^2} - e^{2xy} - y\right)\cdot\frac{1}{x}$$

(11) $$\boxed{\frac{dy}{dx} = e^{2xy-x^2} - \frac{1}{x}e^{2xy} - \frac{y}{x}}$$

Problem.6. Classify the following first-order P.D.E's as either linear, almost-linear, quasi-linear, or nonlinear:

Sol'n:

(a) $\underbrace{x e^{y}}\cdot z_x + \underbrace{x^2 z}\cdot z_y = \underbrace{x\cos(y)}$

$P(x,y,z)\cdot z_x + Q(x,y,z)\cdot z_y = R(x,y,z)$, thus

the given eq'n in (a) is <u>quasi-linear</u> PDE

(b) $x e^{y}\cdot z_x + x^2\cdot z_y = x z \implies$ let me write it as

$\underbrace{(x e^{y})}\cdot z_x + \underbrace{(x^2)}\cdot z_y + \underbrace{(-x)}\cdot z = \underbrace{0}$

$a(x,y)\cdot z_x + b(x,y)\cdot z_y \quad c(x,y)\cdot z = d(x,y)$, hence

the given eq'n in (b) is <u>linear</u> PDE.

(c) $\underbrace{z}\, z_y = \underbrace{5x}$

$Q(x,y,z)\cdot z_y = R(x,y,z)$, thus it is <u>quasi-linear</u> P.D.E.

(d) $\underbrace{x}\cdot \frac{\partial z}{\partial x} + \underbrace{1}\cdot\frac{\partial z}{\partial y} = \underbrace{e^{z}}$

$P(x,y)\cdot z_x + Q(x,y)\cdot z_y = R(x,y,z)$, so the eq'n is <u>almost-linear</u>.

(e) $\underbrace{(x+y)}\, u_x + \underbrace{(y+z)}\, u_z = e^{z}$ (so $u = u(x,y,z)$ at least)

$\underbrace{a(x,y,z)}_{\text{fun'n of indep. v's}}\cdot u_x + \underbrace{c(x,y,z)}_{\text{f'n of only indep. var's}}\cdot u_z = \underbrace{e(x,y,z)}_{\text{f'n of only indep. var's}}$, hence

the given eq'n in (e) is <u>linear</u>.

(f) $x y z_x = \left(\underbrace{(z_y)}^2 - x^2\right)\cdot z$

Because of this $(z_y)^2$, the given eq'n is <u>non-linear</u> P.D.E.